Analog Optical Signal Processing for Astronomical Events

Jennita Rachel A

Overview

Goal

The goal is to create an all-analog system to detect the optical signals, to process those signals with various stages and to classify them.

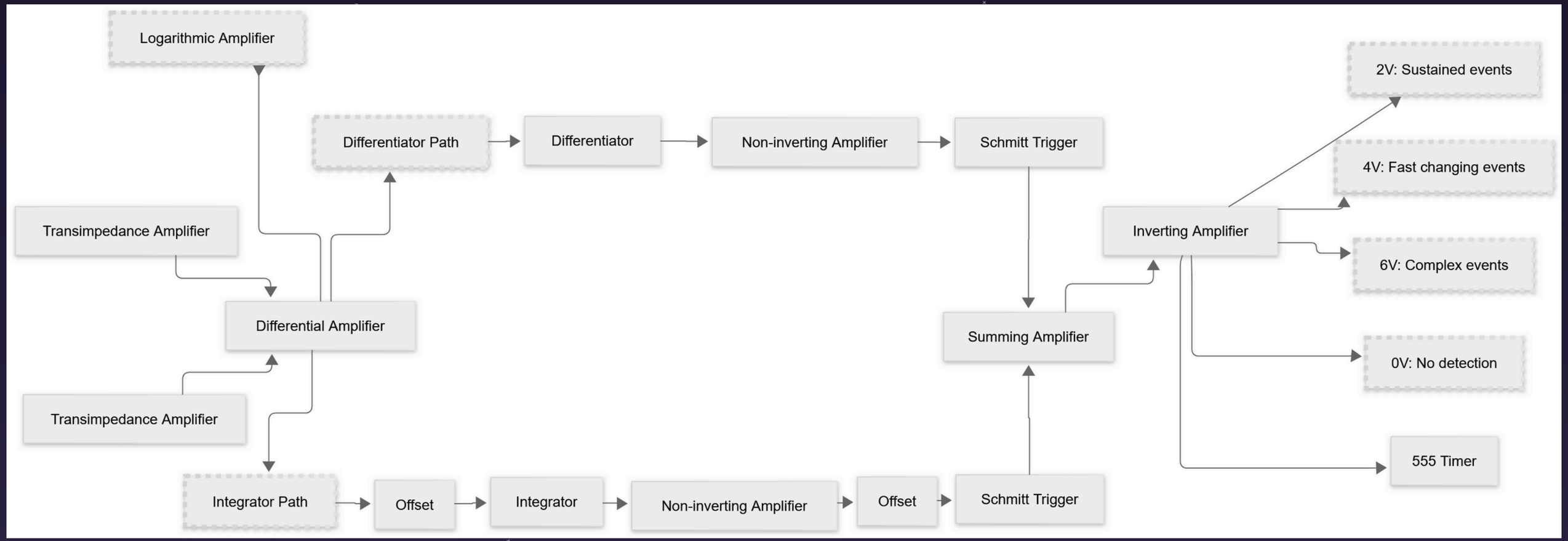
What's included?

- Signal Acquisition
- Noise Rejection
- Parallel Processing
- Event Classification and Pulse counting

Key Features

- Fast/ sustaining/ complex event classification
- Brightness ratio identification
- Count the number of single event

System Flowchart



Input Signal and Noise Rejection

Input

- Optical signal → Photodiode → Transimpedance amplifier
- Used Transimpedance Amplifiers (x2) to convert photodiode current to voltage
- $V_o = -I_i \times R_f$

Differential Amplifier

- Common-mode noise rejection
- Removes light pollution, retains event signal
- $V_o = A_d(V_+ - V_-)$

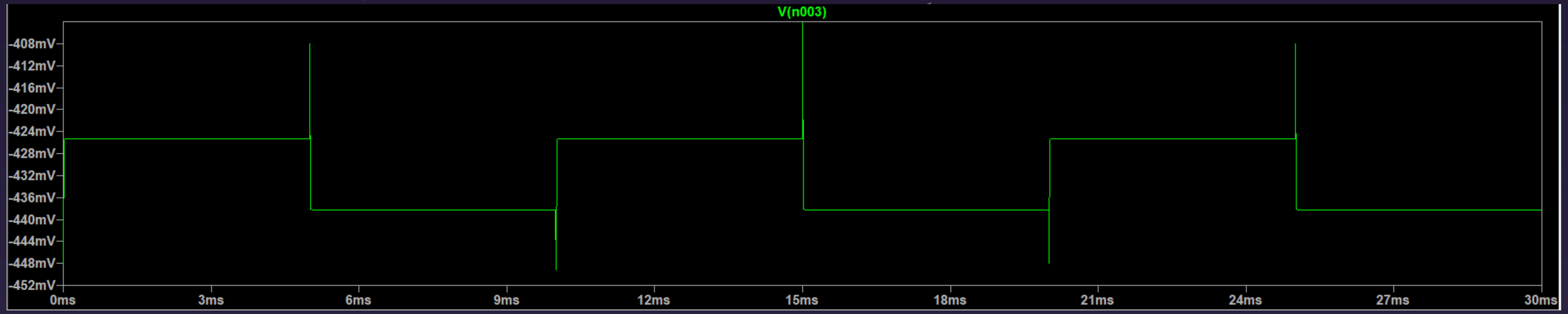
Brightness Ratio

Problem

- Bright Objects - Saturate
- Faint Object - Miss out
- Cannot capture intensity in linear scale

Logarithmic Compression

- Converts high range of signal into manageable range
- $V_o = -V_T \times \ln(V_i / (I_s \times R))$



Differentiator Path

Fast Change Detection

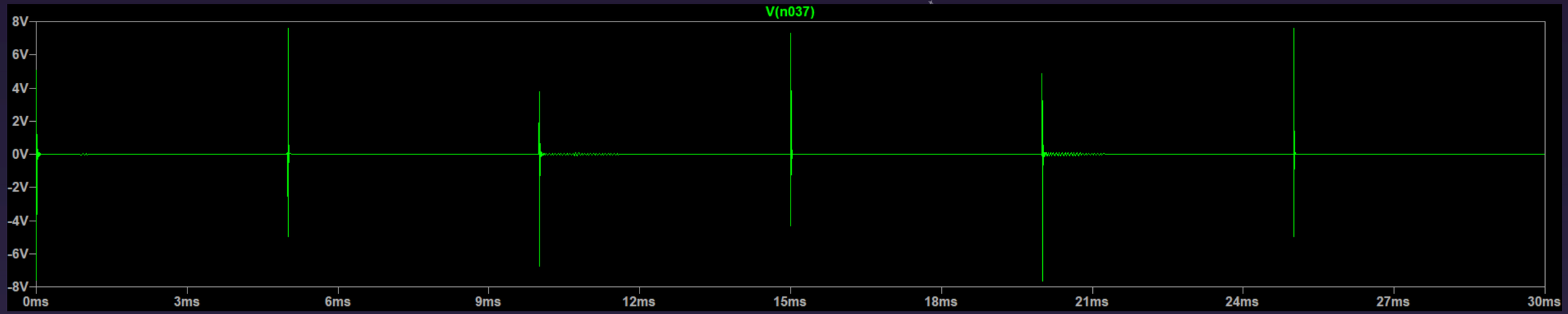
- Differentiator responds to changes faster than 4.7μs
- Meteor streaks, Optical bursts, Fast transients are detected
- **$V_o = -RC \cdot dV_i/dt$**

Detection Threshold

R = 100 kΩ
C = 47 pF

$\tau = R \times C = 100{,}000 \times (47 \times 10^{-12})$ s
= 4.7 μs

As several individual bursts have been observed with features as brief as a few microseconds , the threshold is taken as 4.7 μs



Integrator Path

Slow Change Detection

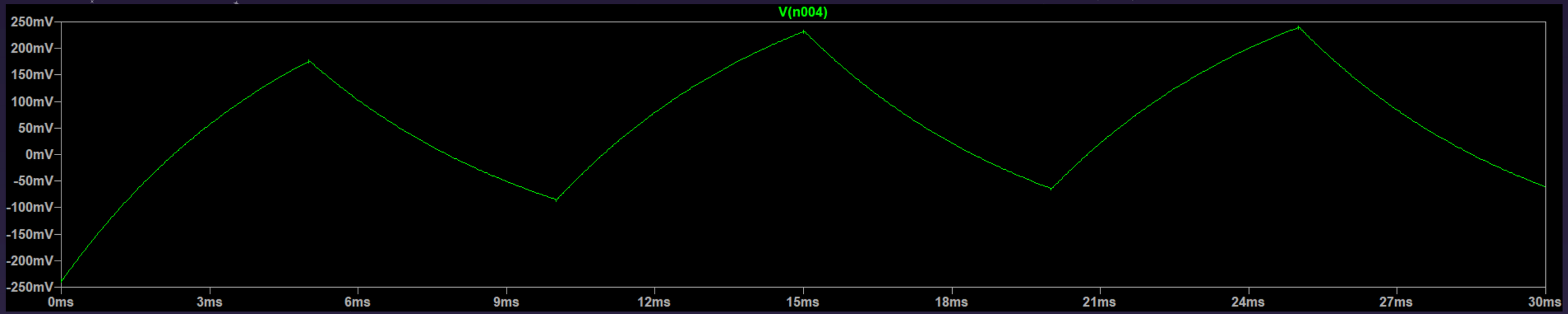
- Integrator responds to accumulation time more than 0.1 s
- Variable stars, Planetary transits, Comet brightening are detected
- **$V_o = -1/RC \int V_i \, dt$**

Detection Threshold

R = 100 kΩ
C = 1μF

$\tau = R \times C = 100{,}000 \times (1 \times 10^{-6})$ s
= 0.1 s

As several sustained events have been observed with features as low as a fraction of a second, so our threshold is taken as 0.1 s



Event Classification

Schmitt Trigger

- **Differentiator path:**

1) Triggers when differentiator output voltage > 1V and output 4V

2) Average fast event has change of 5V change in 10 μs - Using $V_o = -RC \cdot dV_i/dt$ we get Vo to be 2.35V

- **Integrator path:**

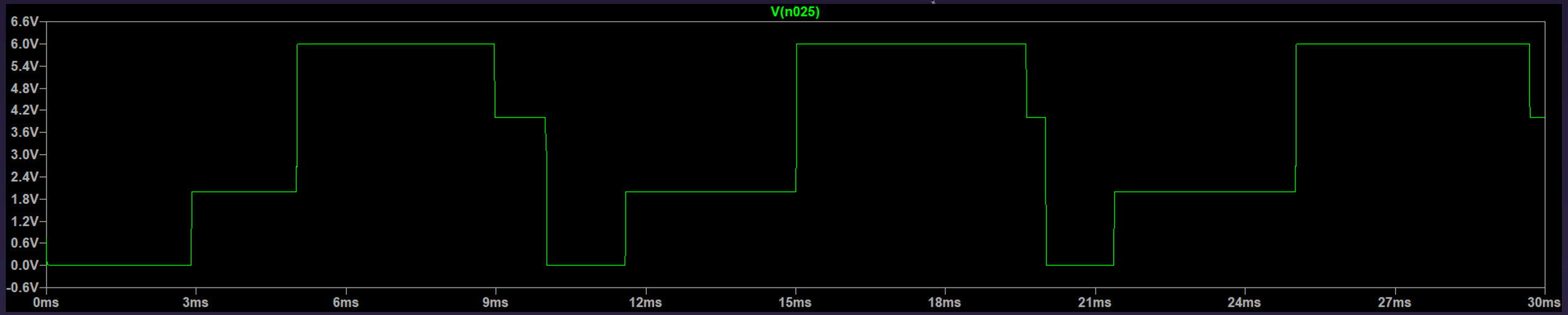
1) Triggers when integrator output voltage > 50 mV and output 2V.

2) Average sustained event has 0.2V maintained for 30ms- Using $V_o = V_o = -1/RC \int V_i \, dt$ we get Vo to be 70 mV

Classification

On summing both paths and inverting:

- 2V - Only Integrator triggered (Sustained Event)
- 4V - Only Differentiator triggered (Fast Event)
- 6V - Both triggered (Complex Event)
- 0V - No triggers (No Significant Event)



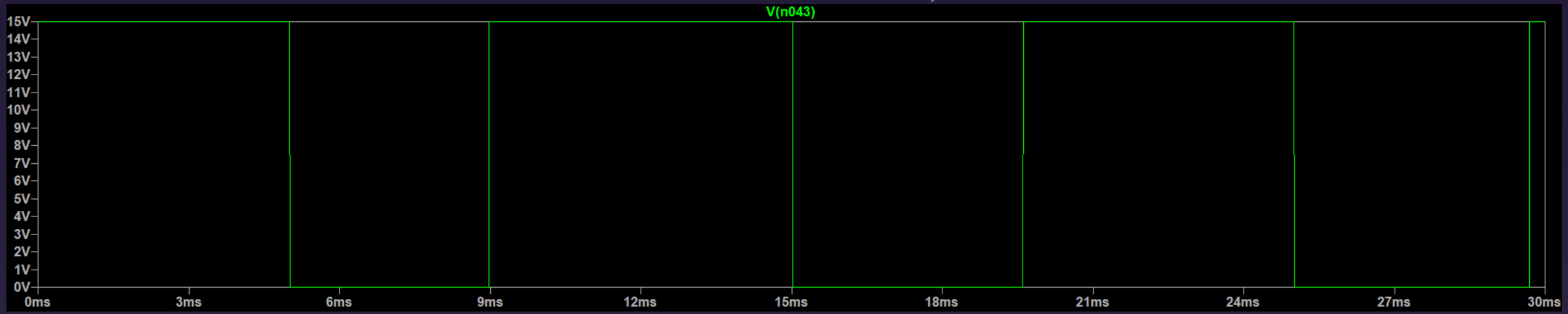
Counting using 555 timer

Working

- 555 compares trigger pin to (1/3) of supply voltage - 15/3V = 5V
- Triggers for <5V
- Either Fast or Slow event, not both, not none
- Pulse width= t = 1.1 × R × C

Output

- R = 3kΩ, C = 1μF, t= 1.1 × 3k× 1μs =3.6ms
- But we observe around 6mV due to CV stabilizing
- Generates equal pulse width with each representing some event has occured but not both
- Total number of pulse gives total number of single event



Conclusion

Application

- Used to distinguish and categorize the events.
- Used for photometric analysis in astronomy.
- Helps in monitoring the celestial changes for scientific research and discovery.

Suggested Improvements

- Adaptive threshold of Schmitt trigger based on sky conditions.
- Automatic offset calibration to prepare the signal for Schmitt Trigger.
- Temperature dependence of most of the elements can be reduced by using FETs.

THAПK YOU!